

Light-modulated low-frequency noise spectroscopy for deep-level defect characterization in Al-rich AlGaN heterostructures

Cite as: Appl. Phys. Lett. **129**, 112108 (2026); doi: [10.1063/5.0349897](https://doi.org/10.1063/5.0349897)

Submitted: 26 June 2026 · Accepted: 31 August 2026 ·

Published Online: 16 September 2026



View Online



Export Citation



CrossMark

Xiulin Xie,¹ , Zhuoya Peng,¹ and Hao Jiang^{1,2,3,a)}

AFFILIATIONS

¹School of Electronics and Information Technology, Sun Yat-sen University, Guangzhou 510006, China

²State Key Laboratory of Optoelectronic Materials and Technologies, Sun Yat-sen University, Guangzhou 510006, China

³Guangdong Engineering Technology R&D Center of Compound Semiconductors and Devices, Sun Yat-sen University, Guangzhou 510006, China

^{a)} Author to whom correspondence should be addressed: stsjiang@mail.sysu.edu.cn

ABSTRACT

Al-rich Al_xGaN ($x > 0.4$) heterostructures are pivotal building blocks for high-voltage power electronic and deep ultraviolet optoelectronic devices, yet deep-level defect characterization in such ultrawide-bandgap semiconductors remains challenging. Here, we demonstrate that light-modulated low-frequency noise (LFN) spectroscopy enables quantitative profiling of both trap density and energy depth. Under intensity-tunable above-bandgap illumination, the dominant $1/f$ noise from carrier number fluctuations yields a trap state density $N_t \sim 10^{18} \text{ cm}^{-3} \text{ eV}^{-1}$, consistent with literature. A distinct ultralow-frequency generation-recombination noise is suppressed under out-of-band excitation with photon energy $> 2.3 \text{ eV}$, matching a 2.3 eV cathodoluminescence peak and suggesting the presence of a trap, with a V_{III} -related deep acceptor as a plausible candidate. Its effective areal trap density is quantified as $1.52 \times 10^{10} \text{ cm}^{-2}$. The technique's validity is further corroborated by reduced N_t in devices with an AlN/GaN interface layer. This work establishes light-modulated LFN as a versatile, device-intrinsic method for deep-trap characterization in ultrawide-bandgap semiconductors, independent of specific device architectures.

Published under an exclusive license by AIP Publishing. <https://doi.org/10.1063/5.0349897>

Al-rich Al_xGaN ($x > 0.4$) heterostructures are foundational to next-generation ultrawide-bandgap (UWBG) semiconductor technologies, enabling high-voltage power electronics^{1–3} and deep ultraviolet optoelectronics.^{4–6} However, deep-level defects, often introduced during epitaxial growth or processing, severely degrade device performance and reliability. Conventional deep-level characterization techniques face fundamental challenges in such material systems. Deep-level transient spectroscopy (DLTS), for instance, relies on thermally stimulated carrier emission, which becomes impractically slow for deep traps in UWBG materials due to exponentially suppressed emission rates.⁷ Deep-level optical spectroscopy (DLOS) and related capacitance-based methods require well-defined rectifying junctions,⁸ such as Schottky diodes or p–n junctions with low leakage, stable barrier characteristics, and measurable depletion capacitance. In high-Al-content AlGa_N, however, such junctions are difficult to fabricate due to the lack of p-type doping, severe Fermi-level pinning, and high bulk/surface defect densities that lead to excessive leakage currents

and poor capacitive response. Consequently, there have been scarcely any quantitative reports on deep-level defect energetics or densities in AlGa_N with Al fractions exceeding 40%, representing a critical knowledge gap.

Beyond thermal activation, optical excitation offers another route to interrogate deep-level states. Above-bandgap (in-band) illumination was shown to activate deep levels associated with defect complexes, attributed to photo-induced changes in their charge state.^{9,10} In contrast, deep sub-bandgap (out-of-band) photons, with energies far below the AlGa_N bandgap (e.g., 1.3–2.9 eV), cannot generate electron–hole pairs but can resonantly excite carriers between discrete energy levels within deep traps, thereby modulating their occupancy and observable noise signatures.

In this work, we introduce light-modulated low-frequency noise (LFN) spectroscopy as an electrically nondestructive, device-intrinsic technique for deep-level defect analysis in Al-rich AlGa_N heterostructures. By using optical intensity as a tunable control parameter,

mimicking gate-like modulation of the two-dimensional electron gas (2DEG) channel, we actively probe defect responses under operational conditions. Through combined in-band and out-of-band illumination, we demonstrate two key capabilities: (i) quantitative extraction of the trap state density (N_t) associated with $1/f$ noise, and (ii) selective identification of specific deep-level defects via suppression of generation-recombination (GR) noise under photon energies exceeding a critical threshold. This dual-mode approach enables comprehensive profiling of both defect density and energy depth, establishing light-modulated LFN as a powerful and broadly applicable tool for UWBG semiconductor characterization.

Two Al-rich AlGaN heterostructures were fabricated for comparative defect analysis: a reference sample and a modified sample with engineered interface quality. Both were grown on c-plane sapphire by metal-organic chemical vapor deposition (MOCVD), featuring a 450 nm AlN buffer, a 70 nm unintentionally doped (uid) Al_{0.53}GaN channel, and a 15 nm uid Al_{0.61}GaN barrier, capped with a 2 nm *in situ* SiN_x layer. The modified structure incorporates 1/1 nm AlN/GaN binary interface layers at the heterointerface to enhance crystallinity and suppress interfacial defects, while otherwise maintaining identical epitaxial design. Cross-sectional schematics of the reference and modified devices are shown in Figs. 1(a) and 1(b), respectively. Interdigitated Ti/Al/Ni/Au (15/80/20/60 nm) Ohmic contacts were formed by electron-beam evaporation followed by rapid thermal annealing at 830 °C for 30 s in N₂. The interdigitated electrode fingers have an active area of $110 \times 105 \mu\text{m}^2$, a length of 100 μm , a width of 3 μm , and a spacing of 5 μm (defining the channel length, L). Further details on growth, processing, and the comprehensive electrical characterization (including 2DEG sheet density and mobility) of these heterostructures are provided in Ref. 11, which details the intentional interface engineering leading to the observed enhancement in 2DEG properties.¹¹

Low-frequency noise (LFN) measurements were performed using a system shown in Fig. 1(c). A Keithley 6517B electrometer supplied the DC bias and simultaneously monitored the channel current under illumination from an intensity-tunable 260 nm light-emitting diode (LED). To probe sub-bandgap defect responses, additional LEDs with center wavelengths of 980, 530, 480, and 430 nm were employed. Current fluctuations were amplified by a low-noise transimpedance preamplifier (SRS SR570) and analyzed with a dynamic signal analyzer (SRS SR785) over the frequency range of 1 Hz–10 kHz. Ultralow-frequency spectra were acquired using the same measurement chain, with the SR785 frequency span reduced to 50 or 100 Hz to provide sufficient frequency resolution. The electrical measurements were typically performed under a drain bias of 5 V.

Figures 2(a) and 2(b) show the current noise power spectral density (S_I) of the reference and modified AlGaN heterostructure devices under variable 260 nm illumination, which controls the 2DEG

channel conductivity via the photogating effect. In both cases, the noise spectra exhibit a $1/f^\gamma$ dependence with $\gamma \approx 1.5$ (reference) and 1.4 (modified). Fits at each optical intensity show only minor variations in γ , with no systematic dependence on I_{ph} or V_{ph} . This deviation from the ideal $1/f$ behavior ($\gamma = 1$) is consistent with the significant alloy disorder and nonuniform trap distribution, which create a broad distribution of trap time constants. The superposition of multiple generation-recombination processes thus results in $\gamma > 1$, a behavior consistent with theoretical descriptions of disordered semiconductor systems¹² and prior observations in III-nitride heterostructures where significant alloy disorder and nonuniform trap distributions are present.^{13,14} However, room-temperature spectral slopes alone cannot uniquely determine the defect-energy distribution or exclude contributions from thermally activated centers such as Fe-related impurities.¹⁵ In our analysis, we directly utilize these experimentally determined γ values for spectral fitting within the carrier number fluctuation (CNF) model. This approach, rather than assuming an ideal $\gamma = 1$, ensures a self-consistent and robust extraction of the trap density N_t , as it inherently accounts for the material's complex trap kinetics.

The modulation of the 2DEG channel by optical intensity can be equivalently regarded as applying a virtual gate voltage through the photogating effect, consistent with the open-circuit voltage theory for HEMTs.¹⁶ This photogating voltage, V_{ph} , physically originates from the splitting of quasi-Fermi levels under illumination. In our self-depleted AlGaN heterostructure,¹⁷ the restored 2DEG under light constitutes a degenerate electron gas with a high density of states. Consequently, the electron quasi-Fermi level shift (ΔE_{Fn}) is negligible. Therefore, V_{ph} is dominated by the shift of the hole quasi-Fermi level (ΔE_{Fp}), induced by accumulated photogenerated holes that screen the built-in polarization field, yielding the expression^{5,17}

$$V_{\text{ph}} = \frac{kT}{q} \ln \left(\frac{\Delta p}{N_v \exp \left(\frac{E_v - E_f}{kT} \right)} + 1 \right), \quad (1)$$

where $\Delta p = \eta_{\text{ext}} P_{\text{inc}} \tau / h\nu d$ represents the concentration of photogenerated holes, with η_{ext} , P_{inc} , τ , $h\nu$, and d denoting the external quantum efficiency, incident optical power density, hole lifetime, photon energy, and thickness, respectively. Here, η_{ext} and τ are treated as approximately constant over the optical-intensity range used in Fig. 2. At very weak or high excitation, trap capture, carrier recombination, and polarization screening can alter the photogating response and the effective τ and η_{ext} .¹⁷ Under the internal polarization field, these photogenerated holes are separated from electrons and swept away from the heterointerface. This separation of charge carriers effectively screens the polarization-induced electric field, which is equivalent to applying a forward gate bias, the V_{ph} , that dynamically recovers and



FIG. 1. Cross-sectional schematics of the (a) reference and (b) modified AlGaN heterostructures. (c) Diagram of the light-modulated low-frequency noise spectroscopy setup.

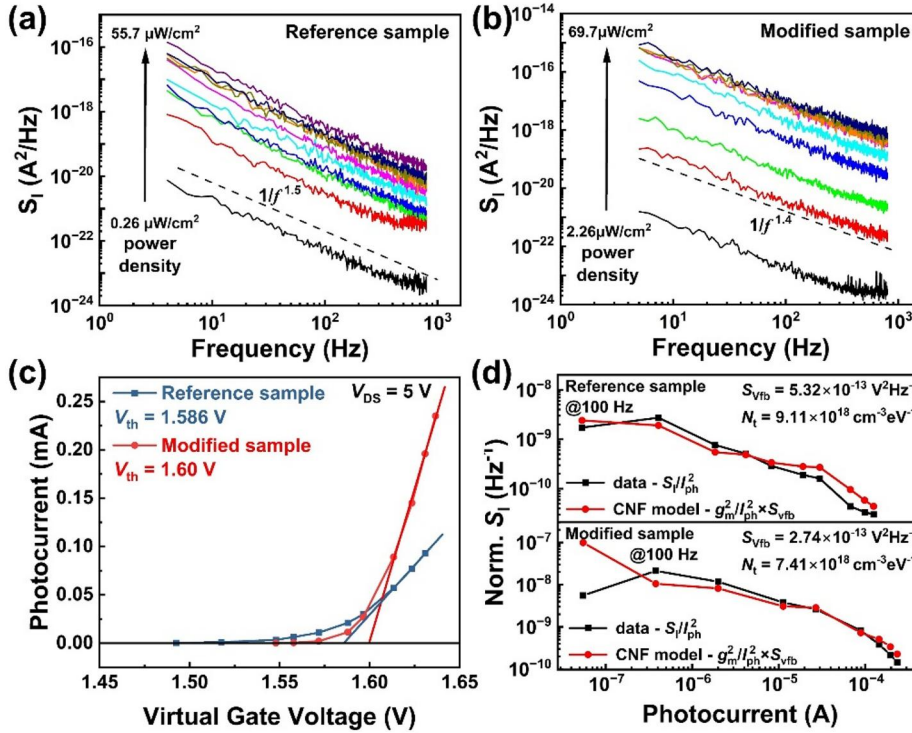


FIG. 2. Frequency-dependent S_I (S_I/I_{ph}^2) under different 260 nm optical intensities for the (a) reference and (b) modified devices. (c) Corresponding I_{ph} - V_{ph} characteristic. (d) Dependence of normalized S_I and $(g_m/I_{ph})^2$ on I_{ph} .

modulates the 2DEG conductivity. This allows us to establish the relationship between photocurrent (I_{ph}) and the equivalent V_{ph} , as plotted in Fig. 2(c). The nearly identical threshold voltages (V_{th}) indicate a similar electrostatic condition to initiate channel conduction in both self-depleted heterostructures. However, the steeper slope in the above-threshold region for the modified sample demonstrates its superior channel properties and more efficient photogating control. Note that the higher g_m in the modified sample arises from the enhanced interface properties. As quantitatively reported in Ref. 11, the binary AlN/GaN interface layer simultaneously increases the 2DEG sheet concentration (from 1.0×10^{12} to 2.5×10^{13} cm⁻²) and electron mobility (from 197 to 268 cm²/V s), a result of modified interfacial polarization and band profile. The observed g_m enhancement under our operating conditions ($V_{DS} = 5$ V, $L = 5$ μm) is explained by long-channel saturation physics. In this regime, g_m is proportional to the product of mobility (μ) and differential capacitance (C_{diff}), i.e., $g_m \propto \mu \cdot C_{diff}$. The enhancement is therefore the result of the combined increases in μ and C_{diff} . The C_{diff} (defined as $q\partial n_s/\partial V_{GS}$) is governed by the geometric barrier capacitance (C_b) in series with the quantum capacitance (C_q) of the 2DEG. The AlN/GaN interface in the modified sample creates a deeper quantum well, enabling a much higher 2DEG density and a large C_q and consequently $C_{diff} \approx C_b$. In contrast, the low-density 2DEG in the reference sample results in a much smaller effective C_q , significantly limiting C_{diff} . Thus, the combined increases in μ and C_{diff} account for the larger of the modified sample.

The extracted g_m and the relationship between I_{ph} and the effective gate bias V_{ph} [Fig. 2(c)] enable a quantitative noise analysis analogous to that in a gated HEMT. Here, scanning the optical power (via

V_{ph}) modulates the quasi-Fermi level to probe trap activity, replacing the conventional gate-voltage sweep. This allows us to apply the CNF model to the normalized noise spectra for trap density extraction. The normalized noise spectral density is then related to the flat-band voltage noise spectral density (S_{Vfb}) by¹⁸

$$S_I/I^2 = (g_m/I)^2 S_{Vfb}. \quad (2)$$

Figure 2(d) plots the normalized noise spectral density (S_I/I^2) vs I_{ph} for both samples over a wide range of photocurrents ($\sim 5 \times 10^{-7}$ to 2×10^{-4} A), covering the operational regime from sub-threshold to above threshold. The excellent agreement with the CNF theory fit ($g_m^2/I_{ph}^2 \times S_{Vfb}$) confirms that the $1/f$ noise in both heterostructures originates from the same physical mechanism, which is the carrier number fluctuations due to trapping/detrapping at interface or near-interface traps. We emphasize that correlated mobility fluctuations are not included in the present analysis. This treatment is consistent with the observation that dielectric and carrier screening strongly reduce the scattering contribution of charged defects located more than a few monolayers away from the interface.¹⁹ Therefore, the consistent application of the CNF model allows for a direct and meaningful comparison of the extracted trap density N_t . Note that the parameter S_{Vfb} obtained from the fit represents an effective flat-band voltage noise spectral density that describes the averaged trap response over the measured photocurrent range. It is related to the trap density N_t by $S_{Vfb} = q^2 kT \lambda N_t / (WLC_b^2 F^2)$, where λ , W , L , and C_b denote tunneling attenuation length, channel width, length, and geometric barrier capacitance, respectively. The value of C_b remains constant during the CNF fitting procedure. When trap-induced charge fluctuations are referred to the input as an equivalent flat-band

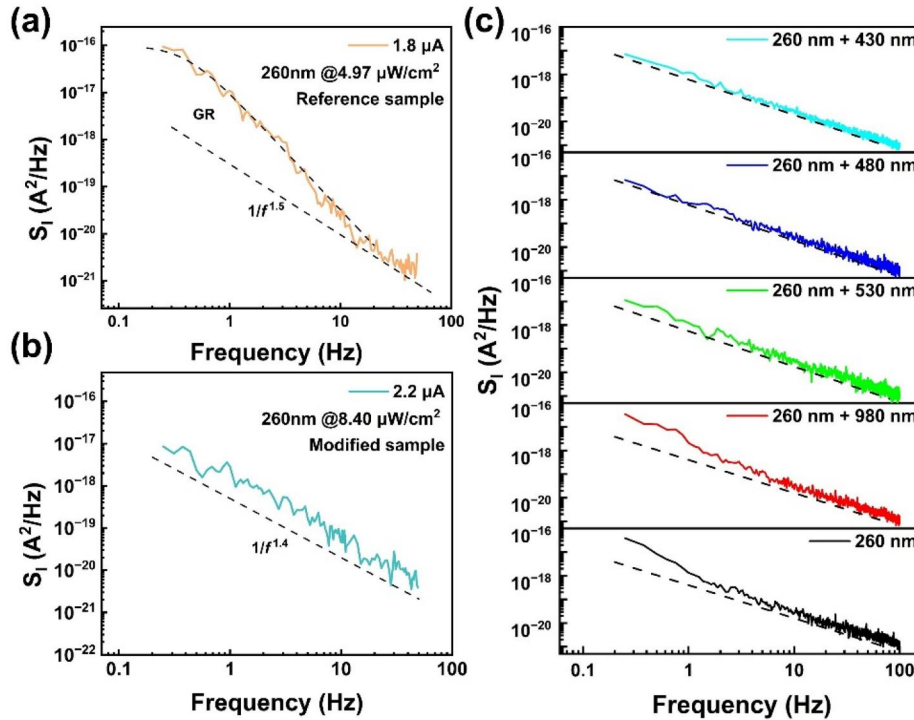


FIG. 3. Ultralow-frequency noise spectra of (a) the reference and (b) the modified AlGaIn heterostructure devices under 260 nm illumination (intensity: 4.97 and 8.40 $\mu\text{W}/\text{cm}^2$). (c) Noise spectra under concurrent out-of-band LED illumination (wavelengths: 980, 530, 480, and 430 nm; intensities: 100–200 $\mu\text{W}/\text{cm}^2$).

voltage, C_q cancels from the N_t expression. Crucially, λ is determined by the WKB approximation: $\lambda = [4\pi(2m^*\Phi_B)^{1/2}/h]^{-1}$, where Φ_B is the tunneling barrier height. For the channel side ($\text{Al}_{0.53}\text{GaIn}$), the potential profile is triangular rather than square. The N_t extracted by this method thus represents the average trap state density within the measured photocurrent range, reflecting the overall concentration of defects within the Fermi level scanning interval.

Based on the 2DEG densities of $1 \times 10^{12} \text{ cm}^{-2}$ (without interlayers) and $2.5 \times 10^{13} \text{ cm}^{-2}$ (with AlN/GaN interlayers), we estimate the effective Φ_B relative to the Fermi level to be in the range of ~ 13 to 100 meV. This yields local tunneling attenuation lengths of $\lambda_{\text{ch}} \approx 1.5 \text{ nm}$ for the reference sample and $\lambda_{\text{ch}} \approx 0.6 \text{ nm}$ for the modified sample. Notably, both values are significantly larger than that of the barrier side ($\lambda_{\text{bar}} \approx 0.46 \text{ nm}$, calculated for the $\text{Al}_{0.61}\text{GaIn}/\text{Al}_{0.53}\text{GaIn}$ band offset $\Delta E_C = 160 \text{ meV}$). Due to the fact that traps in both the barrier and the channel layers are involved in noise generation, using a larger value of λ can more reasonably reflect the contribution of these traps to $S_{V_{\text{bpf}}}$, that is, using λ_{ch} instead of λ_{bar} to calculate N_t in the two samples is more accurate. Consequently, the reference device exhibits $N_t = 9.11 \times 10^{18} \text{ cm}^{-3} \text{ eV}^{-1}$, which is reduced to $7.41 \times 10^{18} \text{ cm}^{-3} \text{ eV}^{-1}$ in the modified sample, extracted from the saturation-regime ($V_{\text{DS}} = 5 \text{ V}$) noise spectra. The CNF model is applicable in this regime because, in our devices, pinch-off is confined near the drain, while the remaining conductive channel is expected to dominate the measured low-frequency noise. The average lateral field ($\sim 10 \text{ kV}/\text{cm}$) is well below the characteristic velocity-saturation field, and the measured g_m also reflects the polarization-dependent channel response at the operating point. To verify this, we performed additional measurements at $V_{\text{DS}} = 0.5 \text{ V}$ (linear regime). The extracted trap densities, $N_t = 9.85 \times 10^{18}$ and

$7.53 \times 10^{18} \text{ cm}^{-3} \text{ eV}^{-1}$, for the reference and modified samples, respectively, are in close agreement with the saturation-regime values (differences $\leq 8\%$), confirming that the extracted N_t is not significantly affected by drain bias.

It is important to note that despite this modest reduction in N_t , the normalized noise [Fig. 2(d)] for the modified sample appears higher at a given current level. The CNF model accounts for it through the enhancement in device g_m . This reduction in N_t is consistent with the improved interface crystallinity afforded by the AlN/GaN interlayers. These values fall within the range reported for AlGaIn/GaN heterojunctions,^{18,20,21} validating the reliability of light-modulated LFN spectroscopy for quantitative defect profiling in UWBG semiconductors.

The investigation of noise behavior extends to the ultralow-frequency range (0.1–50 Hz) to probe deeper defect dynamics. Under constant 260 nm illumination, a clear generation–recombination (GR) noise component is observed in the reference AlGaIn heterostructure, manifesting as a Lorentzian bulge in the $S_I(f)$ spectrum [see Fig. 3(a)]. This Lorentz bulge is most pronounced at a photocurrent of $1.8 \mu\text{A}$ (under $4.97 \mu\text{W}/\text{cm}^2$ illumination) in the reference device, a point lying in the sub-threshold region of its $I_{\text{ph}}-V_{\text{ph}}$ curve [Fig. 2(c)]. This indicates that the quasi-Fermi-level shift at this intensity results in the most intense electron capture and release of the trap level (as discussed below). In contrast, this feature is absent in the modified device incorporating the AlN/GaN interface layer under different light intensities. For comparison, the noise spectrum of the modified device at a comparable photocurrent of $2.2 \mu\text{A}$ shows no such feature [Fig. 3(b)]. This stark difference indicates the presence of a specific trap with a relatively high density and discrete energy level in the reference device, which is effectively suppressed by the interface engineering.

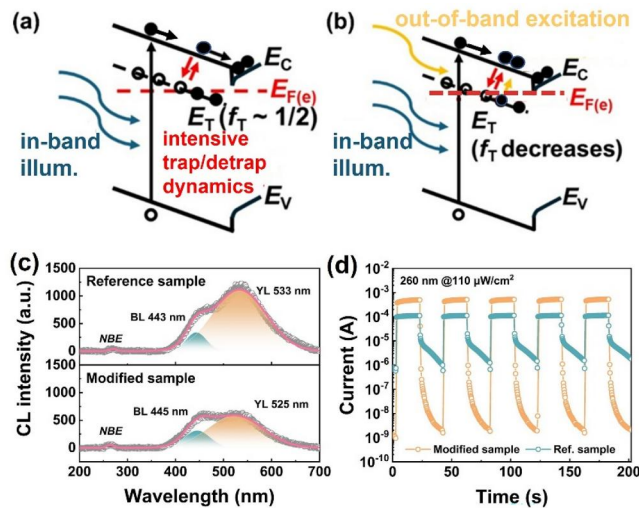


FIG. 4. (a) Schematic energy band diagram of the barrier region under sole 260 nm in-band illumination. (b) Schematic energy band diagram under additional excitation by out-of-band photons ($h\nu > 2.3$ eV). (c) CL spectra of the two wafer samples. (d) Time dependence of photoresponse current of the two heterostructure devices at 5 V bias.

To probe the optical activation energy of this trap, we applied out-of-band excitations while maintaining a constant 260 nm background illumination. As shown in Fig. 3(c), the GR noise amplitude remains unchanged under 980 nm light but is abruptly suppressed at 530 nm and shorter wavelengths, corresponding to a threshold photon energy of ~ 2.3 eV. This phenomenon can be explained by the occupation probability of the trap (f_T) in the GR noise theory.²² The GR noise magnitude is proportional to $f_T(1 - f_T)$, reaching its maximum when $f_T \approx 0.5$. Under specific 260 nm illumination, the Fermi level aligns such that f_T is near 0.5, resulting in observable GR noise. When photons with energy > 2.3 eV are introduced, they photoionize the traps, exciting electrons from the defect level to the conduction band. This process reduces f_T , driving it away from 0.5 and thereby suppressing the GR noise amplitude. The corresponding mechanism is schematically illustrated in the energy band diagrams in Figs. 4(a) and 4(b), which depict the shift of the quasi-Fermi level [$E_{F(e)}$] away from the trap state upon out-of-band excitation.

As the GR noise exhibits a Lorentzian line shape with a plateau magnitude A_{SI} and a corner frequency $f_0 = 0.4$ Hz, the effective areal trap density (N_{eff}) is extracted using these parameters by the following expressions:^{23,24}

$$A_{SV} = \frac{q^2 N_{eff}}{WLC_b^2 2\pi f_0}, \quad A_{SV} = A_{SI} / g_m^2. \quad (3)$$

The value of N_{eff} was calculated to be $1.52 \times 10^{10} \text{ cm}^{-2}$. Notably, the well-defined single corner frequency f_0 yielding this quantitative result corresponds to a time constant $\tau = (2\pi f_0)^{-1} \approx 400$ ms. This value agrees remarkably well with the slow decay component ($\tau_2 = 431$ ms) observed in the persistent photoconductivity measurements (to be discussed in detail later), reinforcing the conclusion that the GR noise originates from a specific, dominant deep-level defect rather than a distribution of traps.

Figure 4(c) displays the cathodoluminescence (CL) of the reference and modified samples, measured at an acceleration voltage of 5 kV to ensure the probing depth correlates with the noise-generating region. At this voltage, the electron interaction volume is confined primarily within the top ~ 100 to 200 nm, effectively encompassing the 15 nm barrier layer and the underlying channel layer (~ 70 nm) where the 2DEG and interface states are located. The spectra reveal two dominant defect-related emissions at 440 nm (2.8 eV) and 530 nm (2.3 eV). Whereas the integrated area of the 2.8 eV peak remains almost unchanged in the modified sample relative to the reference, the 2.3 eV peak decreases by over half. The GR noise is attributed to this deep-level defect based on the correspondence between the decrease in the 2.3 eV CL peak and the suppression of GR noise under out-of-band illumination. Based on the energy position and literature reports, this defect is associated with transitions involving shallow donors and deep acceptors, likely related to V_{III}^{3-} vacancies.^{25,26} Carbon-related deep acceptors should also be considered as possible contributors.²⁷ Furthermore, the impact of this defect on device dynamics is confirmed by the photoresponse decay characteristics [see Fig. 4(d)]. The decay curves were fitted using a bi-exponential model given by $[I(t) = A_1 \exp(-t/\tau_1) + A_2 \exp(-t/\tau_2) + I_{dark}]$ for the reference device and a mono-exponential model given by $[I(t) = A_1 \exp(-t/\tau_1) + I_{dark}]$ for the modified one. Accordingly, the reference device exhibits persistent photoconductivity with slow decay components (6.2 and 431 ms), whereas the modified device shows a single fast decay component (0.043 ms). This directly demonstrates that the deep-level defects responsible for the GR noise also govern the carrier recombination kinetics, limiting the device's response speed.

In summary, we establish light-modulated low-frequency noise spectroscopy as a device-intrinsic method for deep-level defect characterization in ultrawide-bandgap AlGaIn heterostructures. The technique enables simultaneous quantification of trap state density ($N_t \sim 10^{18} \text{ cm}^{-3} \text{ eV}^{-1}$) and energy-resolved identification of specific defects, demonstrated by the suppression of generation-recombination noise under sub-bandgap illumination with photon energy > 2.3 eV, with the threshold matching a cathodoluminescence peak and suggesting an interface-sensitive deep-level defect, potentially associated with V_{III} -related deep acceptors. Unlike conventional thermal or junction-dependent techniques, this approach operates at room temperature, requires no special device architecture, and directly probes defects under operational conditions. It thus provides a powerful electrical route to deep-trap fingerprinting in next-generation wide-bandgap semiconductors.

This work was supported by the Key Program of the National Natural Science Foundation of China (Grant No. 62534003).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Xiulin Xie: Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Zhuoya Peng:** Formal analysis (equal); Investigation (equal); Writing – review &

editing (equal). **Hao Jiang**: Conceptualization (lead); Funding acquisition (lead); Project administration (lead); Resources (lead); Supervision (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- ¹S. Mukhopadhyay, C. Liu, J. H. Chen, M. T. Alam, S. Sanyal, R. X. Bai, G. Y. Wang, C. Gupta, and S. S. Pasayat, *Crystals* **13**(10), 1456 (2023).
- ²H. Tokuda, M. Hatano, N. Yafune, S. Hashimoto, K. Akita, Y. Yamamoto, and M. Kuzuhara, *Appl. Phys. Express* **3**(12), 121003 (2010).
- ³A. Yamada, J. Yaita, N. Nakamura, and J. Kotani, *J. Cryst. Growth* **560–561**, 126046 (2021).
- ⁴Q. Cai, H. You, H. Guo, J. Wang, B. Liu, Z. Xie, D. Chen, H. Lu, Y. Zheng, and R. Zhang, *Light* **10**(1), 94 (2021).
- ⁵J. Lu, Z. Lv, X. Qiu, S. Lai, and H. Jiang, *Photonics Res.* **10**(9), 2229 (2022).
- ⁶P. Reddy, M. H. Breckenridge, Q. Guo, A. Klump, D. Khachariya, S. Pavlidis, W. Mecouch, S. Mita, B. Moody, J. Tweedie, R. Kirste, E. Kohn, R. Collazo, and Z. Sitar, *Appl. Phys. Lett.* **116**(8), 081101 (2020).
- ⁷M. Buffolo, A. Caria, F. Piva, N. Roccato, C. Casu, C. De Santi, N. Trivellin, G. Meneghesso, E. Zanoni, and M. Meneghini, *Phys. Status Solidi A* **219**(8), 2100727 (2022).
- ⁸N. Roccato, F. Piva, C. De Santi, M. Buffolo, M. Fregolent, M. Pilati, N. Susilo, D. H. Vidal, A. Muhin, L. Sulmoni, T. Wernicke, M. Kneissl, G. Meneghesso, E. Zanoni, and M. Meneghini, *Appl. Phys. Lett.* **122**(16), 161105 (2023).
- ⁹G. Alfieri, L. Knoll, L. Kranz, and V. Sundaramoorthy, *J. Appl. Phys.* **123**(17), 175304 (2018).
- ¹⁰T. Owa, J. Miura, F. Inamura, T. Ikuta, K. Maehashi, and K. Ikushima, *Jpn. J. Appl. Phys.* **64**(12), 12SP04 (2025).
- ¹¹Z. Y. Peng, Z. S. Lv, S. Q. Yang, Y. Yin, J. B. Lu, and H. Jiang, *IEEE Electron Device Lett.* **46**(10), 1725 (2025).
- ¹²M. Haartman and M. Östling, *Low-Frequency Noise in Advanced MOS Devices* (Springer Science & Business Media, Dordrecht, The Netherlands, 2007), p. 68.
- ¹³J. Chen, Y. S. Puzyrev, R. Jiang, E. X. Zhang, M. W. McCurdy, D. M. Fleetwood, R. D. Schrimpf, S. T. Pantelides, A. R. Arehart, S. A. Ringel, P. Saunier, and C. Lee, *IEEE Trans. Nucl. Sci.* **62**(6), 2423 (2015).
- ¹⁴P. F. Wang, X. Li, E. Zhang, R. Jiang, M. W. McCurdy, B. S. Poling, E. R. Heller, R. D. Schrimpf, and D. M. Fleetwood, *IEEE Trans. Device Mater. Reliab.* **20**(2), 420 (2020).
- ¹⁵D. M. Fleetwood, X. Li, E. X. Zhang, R. D. Schrimpf, and S. T. Pantelides, *IEEE Trans. Electron Devices* **71**(2), 1024 (2024).
- ¹⁶Z. H. Zaidi and P. A. Houston, *IEEE Trans. Electron Devices* **60**(9), 2776 (2013).
- ¹⁷J. Lu, Z. Lv, and H. Jiang, *Photonics Res.* **11**(7), 1217 (2023).
- ¹⁸K.-S. Im, U. Choi, M. Kim, J. Choi, H.-S. Kim, H.-Y. Cha, S. J. An, and O. Nam, *Appl. Phys. Lett.* **120**(1), 012102 (2022).
- ¹⁹D. M. Fleetwood, *Appl. Phys. Lett.* **122**(17), 173504 (2023).
- ²⁰C. Liu, Y. Chen, Y. Xie, H. Liu, and Z. Cai, *Micromachines* **15**(9), 1100 (2024).
- ²¹W. Amir, J.-W. Shin, K.-Y. Shin, J.-M. Kim, C.-Y. Cho, K.-H. Park, T. Hoshi, T. Tsutsumi, H. Sugiyama, H. Matsuzaki, and T.-W. Kim, *Sci. Rep.* **11**(1), 22401 (2021).
- ²²F. Grüneis, *Physica A* **568**, 125748 (2021).
- ²³D. Boudier, B. Cretu, E. Simoen, R. Carin, A. Veloso, N. Collaert, and A. Thean, *Solid-State Electron.* **128**, 109 (2017).
- ²⁴K. Takakura, V. Putcha, E. Simoen, A. R. Alian, U. Peralagu, N. Waldron, B. Parvais, and N. Collaert, *IEEE Trans. Electron Devices* **67**(8), 3062 (2020).
- ²⁵X. J. Qiu, Z. S. Lv, Y. Y. He, Z. S. Wu, and H. Jiang, *Superlattices Microstruct.* **131**, 59 (2019).
- ²⁶Y. S. Puzyrev, T. Roy, M. Beck, B. R. Tuttle, R. D. Schrimpf, D. M. Fleetwood, and S. T. Pantelides, *J. Appl. Phys.* **109**(3), 034501 (2011).
- ²⁷W. R. Wampler, A. M. Armstrong, and G. Vizkelethy, *J. Appl. Phys.* **132**(9), 095702 (2022).